**Suggested problems:** 1-5

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**Problem 1:** Suppose that $X \sim \mathcal{U}nif(a, b)$ is a **discrete** uniform random variable. Compute the expectation $E(X)$.

**Problem 2:** Suppose $X \sim \mathcal{B}er(p)$. Compute the expectation $E(X)$.

**Problem 3:** (Example 4.1.2) A stock purchased for \$18 is worth $18 + X$ a year later, where $X$ is a discrete random variable with p.f. given by

$$f(x) = \begin{cases} 0.1 & : x = -2, \\ 0.4 & : x = 0, \\ 0.3 & : x = 1, \\ 0.2 & : x = 4, \\ 0 & : \text{else.} \end{cases}$$

(a) Compute the expected value $E(18 + X)$ of the stock a year later.

(b) Suppose that a person is deciding between investing their \$18 in the stock, or a savings account earning 5% annual interest. Which of the two investments would they choose if they were risk averse? Risk seeking? Risk neutral?

**Problem 4:** Suppose that $X \sim \mathcal{U}nif(a, b)$ is a **continuous** uniform random variable. Compute the expectation $E(X)$.

**Problem 5:** (Example 4.1.6) An appliance has a maximum lifetime of one year. The time $X$ until it fails is a random variable with p.d.f. given by

$$f(x) = \begin{cases} 2x & : 0 < x < 1, \\ 0 & : \text{else.} \end{cases}$$

Compute the average time till failure.

**Problem 6:** (Example 4.1.11) Suppose that $X$ is a continuous random variable with p.d.f. given by

$$f(x) = \begin{cases} 3x^2 & : 0 < x < 1, \\ 0 & : \text{else.} \end{cases}$$

Compute the expectation of $Y = 1/X$.